Control work 2 · Pythagoras and trigonometry

The Chapter II assessment, and the four errors it produces every year.

LESSON 32 | 1 × 40 MIN | GEOMETRY 8, ТЕМЫ 29–30 | STAGE 9 · UNIT 5 CHECK

LESSON CLOCK

2' | 36' | 2'

Setting up	2 min
The paper	36 min
Collect in	2 min

LEARNING OBJECTIVES

- Assess Pythagoras, the four ratios, the identity, special angles and solving triangles.
- Produce a labelled figure and a stated reason for every answer.
- Diagnose each lost mark by error type.

TEXTBOOK

National	pp. 67–68
Cambridge	Learner's Book pp. 124–126
Workbook	Workbook 5.5

01 Explanation

The paper — 40 minutes, 7 tasks, 14 marks

TASK	TOPIC	SOURCE
1	Find a side by Pythagoras	Тема 17
2	Test a triangle for a right angle	Тема 18
3	Find the four ratios from the sides	Тема 15
4	Use the basic identity	Темы 20–21
5	Exact value for 30°, 45° or 60°	Тема 23
6	Solve a right triangle	Темы 25–26
7	A practical height or distance problem	Тема 28

Say before the paper starts: a numerical answer with no figure and no reason earns one mark out of two. Round to one decimal place unless an exact value is asked for.

Work on mistakes — the four errors

WHAT GOES WRONG

1. **Added where a subtraction was needed** — using $c^2 = a^2 + b^2$ when looking for a leg.
2. **Opposite and adjacent swapped** — sine used where cosine was needed.
3. **The calculator in radian mode**, giving a negative or absurd ratio.
4. **Multiplied instead of divided** when going from a leg to the hypotenuse.

Each has a one-second check: the hypotenuse must be the longest side; a sine must be under 1; the answer must be sensible on the figure. The Hard set below is built from these four.

02 Worked examples

EXAMPLE 1 **Find the error: hypotenuse 13, one leg 5, so the other leg is $\sqrt{169 + 25} \approx 13.9$**

1. The answer is longer than the hypotenuse — impossible.
   That check alone catches it.
2. Looking for a **leg**, so subtract.
3. $b = \sqrt{169 - 25} = \sqrt{144} = 12$

ANSWER

12

EXAMPLE 2 **Find the error: a leg is 6 and the angle opposite it is 30°, so the hypotenuse is $6 \cdot \sin 30° = 3$**

1. The hypotenuse came out shorter than the leg — impossible.

2. $\sin 30° = \frac{a}{c}$, so $c = \frac{a}{\sin 30°}$

   Going leg → hypotenuse means dividing.

3. $c = \frac{6}{0.5} = 12$

ANSWER

12

03 Interactive model

In the mistakes lesson: show the wrong working, take a vote on the error type, then reveal.

Diagnose the error

Claimed: hypotenuse 13, leg 5, other leg $\sqrt{169 + 25} \approx 13.9$

1. A leg cannot be longer than the hypotenuse.

The instant check.

2. To find a leg, subtract:

$b^2 = c^2 - a^2$.

3. $b = \sqrt{144} = 12$

And 5, 12, 13 is a familiar triple.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Solve a triangle	Uchburchakni yechish	Решить треугольник
Figure	Chizma	Чертёж
Reason	Asos	Обоснование
Round to	Yaxlitlash	Округлить до
Error	Xato	Ошибка
Exact value	Aniq qiymat	Точное значение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Hypotenuse 25, one leg 7. The other leg is:

- 24
- 26
- 32
- 18

Given the hypotenuse and the angle, the adjacent leg is:

- $c \sin \alpha$
- $c \cos \alpha$
- $\dfrac{c}{\cos \alpha}$
- $c \tan \alpha$

A calculator returns $sin\ 40 = 0.745$. This is:

- correct
- radian mode
- impossible
- a rounding error

A leg is 9 and the angle opposite is 30°. The hypotenuse is:

- 4.5
- 18
- $9\sqrt{3}$
- 15

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. ***Variant 1, task 1.*** *Legs 12 and 16. Find the hypotenuse.*

   ANSWER 20

2. ***Task 2.*** *Is 7, 24, 25 right-angled?*

   ANSWER Yes — $49 + 576 = 625$.

3. ***Task 3.*** $a = 6, b = 8, c = 10$. *Find* $\sin\alpha$ *and* $\cos\alpha$.

   ANSWER 0.6 and 0.8

4. ***Task 4.*** $\sin\alpha = 0.6$. *Find* $\cos\alpha$.

   ANSWER 0.8

5. ***Task 5.*** *Find* $\sin 60°$ *exactly.*

   ANSWER $\frac{\sqrt{3}}{2}$

6. ***Task 6.*** $c = 10, \alpha = 30°$. *Find both legs.*

   ANSWER 5 and $5\sqrt{3}$

7. ***Task 7.*** *A ladder 10 m long has its foot 6 m from a wall. How high does it reach?*

   ANSWER 8 m

Medium · the standard question

7 PROBLEMS

1. ***Variant 2, task 1.*** *Hypotenuse 26, one leg 24. Find the other leg.*

   ANSWER 10

2. ***Task 2.*** *Classify the triangle with sides 6, 7, 10.*

   ANSWER $36 + 49 = 85 < 100$ — obtuse.

3. ***Task 3.*** $a = 9$, $b = 12$, $c = 15$. *Find* $\tan\alpha$ *and* $\cot\alpha$.

   ANSWER 0.75 and $\frac{4}{3}$

4. ***Task 4.*** $\cos\alpha = \frac{7}{25}$. *Find* $\sin\alpha$.

   ANSWER $\frac{24}{25}$

5. ***Task 5.*** *Find* $\tan 30°$ *exactly.*

   ANSWER $\frac{\sqrt{3}}{3}$

6. ***Task 6.*** $a = 5$, $b = 12$. *Solve the triangle.*

   ANSWER $c = 13$, $\alpha \approx 22.6°$, $\beta \approx 67.4°$

7. ***Task 7.*** *From 50 m away the angle of elevation of a tower top is 40°. Find its height.*

   ANSWER ≈ 42.0 m

Hard · stretch and reason

7 PROBLEMS

1. *Find the error: hypotenuse 13, leg 5, other leg* $\sqrt{169 + 25}$

   ANSWER Added instead of subtracted. Correct: *12*.

2. *Find the error:* $c = 20$, $\alpha = 35°$, *adjacent leg* $= 20 \sin 35°$

   ANSWER Sine uses the opposite leg. Correct: $20 \cos 35° \approx 16.4$.

3. *Find the error:* $\sin 37° = -0.6435$

   ANSWER The calculator is in radian mode. Correct: ≈ 0.602.

4. *Find the error: leg 6 opposite 30°, so* $c = 6 \sin 30° = 3$

   ANSWER Leg → hypotenuse means dividing. Correct: *12*.

5. *Find the error:* $\sin \alpha = 1.25$ *for an acute angle*

   ANSWER Impossible — a sine is under 1. A side has been divided the wrong way round.

6. *Find the error: "sides 5, 12, 13 give* $\sin \alpha = \frac{13}{5}$*"*

   ANSWER The hypotenuse belongs below. Correct: $\sin \alpha = \frac{5}{13}$.

7. *Find the error: "*$\tan 45° = 0$*, since the legs are equal"*

   ANSWER Equal legs give $\frac{a}{a} = 1$, not 0.

07 Homework

After the control work

Re-solve every task you lost marks on. Quarter III opens with the coordinate method.

1. *Re-solve every task you lost marks on, with a labelled figure and a stated reason.*
2. *Write out the four errors from the Hard set in your own words.*

3. *Learn the table of exact values for 30°, 45° and 60° — Quarter III assumes it.*